



Fig. 8: Atmel STK600

on the STK600 main board dependent on the device. The kit offers access to all device pins, and several useful hardware functions such as pushbuttons, LEDs and data Flash to create a complete system for prototyping and testing new designs.

Microchip's PIC18 Explorer board. This low-cost demo board is used for evaluation of Microchip's PIC18 MCU families. PIC18 is the highest-performing 8-bit architecture and has three different families: standard PIC18, PIC18

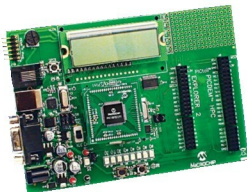


Fig. 9: PIC18 Explorer board

J-series and PIC18 K-series. This single development board supports dozens of general-purpose PIC18 families using various processor plug-in modules. PICtail daughter boards enable many different accessory boards to connect to PIC18 Explorer board for a flexible and complete development environment. It conveniently connects to MPLAB PICkit 3, ICD 3 or REAL ICE for in-circuit programming and debugging. Other features include alphanumeric LCD display, USB interface for USB to RS232 communication, 25LC256 SPI EEPROM, crystal oscillator, potentiometer (connected to

10-bit A/D, analogue input channel), analogue-output temperature sensor, LEDs and RS-232 port.

Renesas' RDK for RX63N (Part No. YRDKRX63N). The Renesas demonstration kit (RDK) for RX63N (32-bit MCU) is an evaluation and demonstration tool for Renesas

RX63N microcontrollers. The goal is to provide the user with a powerful debug and demonstration platform targeted at common applications. The board also provides a useful platform for evaluating the Renesas suite of development tools for coding and debugging, using the High-performance Embedded Workshop (HEW) IDE as well as programming the device using the on-board SEGGER J-Link JTAG debugger. The features are as follows:

1. 32-bit, 100Mhz RX63N MCU @ 165 DMIPS
2. 10/100 Ethernet
3. USB host and device ports
4. 3-axis accelerometer from Analog Devices
5. 128MB PCM serial Flash from Micron
6. Stereo audio output and on-board speaker
7. Analog Devices silicon microphone input
8. Temperature sensor from Analog Devices
9. User pot and three switches
10. Micro SD card slot
11. 96x64 graphics LCD from Okaya
12. Expansion headers
13. Dual Pmod connectors
14. LED simulated motor phases
15. USB or externally powered
16. Total Phase Beagle I2C/SPI debug interface
17. Built-in SEGGER J-Link debugger
18. Myriad of software, code examples, and demos

The Renesas demonstration kit for RX63N is also the hardware platform that is included in a special RX63N-specific edition of Micrium's book on the μ C/OS-III Real Time Kernel. The book and board combo targets serious

Microcontroller boards from other manufacturers

A lot of other manufacturers are also coming up with budget-friendly good-quality development boards. Raspberry Pi and Arduino are the top



examples of the boards that are not really the official boards but have become highly popular. This has motivated a lot of other manufactures also to design and develop budget-friendly development boards. KitsnSpares recently came up with boards like Mango Pi and Guava Pi that are compact, good quality and budget friendly.

Mango Pi is a compact PIC development board with high-end features and budget-friendly price tag. The board is developed and tested by EFY LAB. The main features of the board are as follow:

1. Budget-friendly 8-bit PIC microcontroller (16F877A) development board
2. Can program up to 659 other PIC controllers
3. Easy to test and debug programs with on-board LED and LCD interface
4. On-board microcontroller can be programmed with USB directly without the need of external programmer

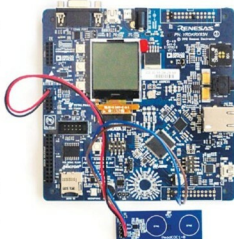


Fig. 10: Renesas' YRDKRX63N board

programmers and college students who need to understand and create programs based on the time-saving features of a real-time kernel. ●

The author is a technical editor at EFY